Exterior Lights (Mod 6/2058) – General Data

1. General

The exterior lighting system consists of position lights, landing lights, a red beacon light, and white strobe lights. All exterior lights are powered from the left or right DC buses. A taxi light is supplied as standard equipment on all aircraft except floatplanes.

2. Position Lights

The position (navigation) lights consist of red and green wing tip lights and a white taillight. The lights are controlled by a switch labelled POSITION on the left sub-panel console. The circuit is powered from the left DC bus and protected by a POSITION LIGHTS (C17) circuit breaker on the main circuit breaker panel. (Refer to 24-30-00, DC Generation – General Data).

3. Landing Lights

Two high intensity discharge (xenon) landing lights are installed with independent power supplies, one in each wing leading edge outboard of the engine (Pre Mod 6/2285). Aircraft incorporating Post Mod 6/2285 have two LED Landing Lights installed, one in each wing leading edge outboard of the engine. The lights are controlled by two switches labelled LANDING LIGHTS that are located on the start panel directly above the left pilot windshield. The STEADY switch activates them continuously. The PULSE switch activates the lights alternately by manual selection, or triggered by the TCAS system (refer to 34-43-00, Traffic Alert and Collision Avoidance System I (TCAS I) – General Data, and 34-43-01, Traffic Alert and Collision Avoidance System II (TCAS II) (Mod S.O.O. 6219) – General data). The circuits are powered from the left and right DC buses and protected by LANDING LIGHT L (C16) and LANDING LIGHT R (D16) circuit breakers on the main circuit breaker panel, capable of pulsing. (Refer to 24-30-00, DC Generation – General Data).

4. Wing Tip Strobe Lights and Beacon Light

Two white strobe type lights are part of the position light assembly installed in each wing tip. A red beacon light is mounted on the top of the vertical stabilizer. The white strobe lights are controlled by a switch labelled STROBE on the lower left pilot sub-panel; the circuit is powered from the left DC bus and is protected by the STROBE LIGHTS (D15) circuit breaker on the main circuit breaker panel. (Refer to 24-30-00, DC Generation – General Data).

The red beacon light is controlled by a switch labelled BEACON on the lower left pilot sub-panel; the circuit is powered from the left DC bus and is protected by the BEACON LIGHT (C15) circuit breaker on the main circuit breaker panel.

5. Taxi Light

Aircraft incorporating S.O.O. 6441 have two incandescent taxi lights installed on the nose wheel fork. The incandescent lights are controlled by the TAXI switch on the lower left pilot sub-panel and the circuits are powered from the right DC bus and protected by TAXI LIGHT L (B16) and TAXI LIGHT R (B14) circuit breakers.

Aircraft incorporating S.O.O. 6442 have two LED taxi lights installed on the nose wheel fork. The taxi lights are controlled by the TAXI switch on the lower left pilot sub-panel and the circuits are powered from the right DC bus and protected by the TAXI LIGHT L (B16) and TAXI LIGHT R (B14) circuit breakers. S.O.O. 6441 and S.O.O. 6442 allow for the operator to easily switch between incandescent and LED taxi lights.

A LED taxi light is installed on the nose wheel fork. The light is controlled by a switch labelled TAXI on the lower left pilot sub-panel and the circuit is powered from the right DC bus and protected by a TAXI LIGHT (B16) circuit breaker on the main circuit breaker panel. The taxi light is not installed on floatplanes. (Refer to 24-30-00, DC Generation – General Data). On aircraft incorporating S.O.O. 6440, a second LED taxi light is installed on the nose wheel fork.

6. Wing Inspection Lights (Mod 6/2028 or Mod 6/2140)

Two 40 watt incandescent wing inspection lights are provided (Pre Mod 6/2140), one on the outboard side of each engine nacelle. Aircraft incorporating Mod 6/2140 have two 6.5 watt LED wing inspection lights, one on the outboard side of the each engine nacelle. The lights illuminate the leading edges of the wings and are controlled by a switch labelled WING INSP on the lower left pilot sub-panel. The circuit is powered from the right DC bus and protected by a WING INSPECTION LIGHT (D17) circuit breaker on the main circuit breaker panel. (Refer to 24-30-00, DC Generation – General Data).

7. Adjustment/Test

A. Operational Test Landing Lights

1) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

2) Switch LANDING LIGHTS STEADY switch to ON and check that left and right landing light comes on.

3) Switch the LANDING LIGHTS PULSE switch to PULSE and check that the left and right landing lights alternate illuminating. Switch the LANDING LIGHTS PULSE switch to OFF. Switch the LANDING LIGHTS STEADY switch to OFF.

Note For testing of the TCAS triggered pulse function refer to 34-43-00, Traffic Alert and Collision Avoidance System I (TCAS I) – General Data and 34-43-01, Traffic Alert and Collision Avoidance System II (TCAS II) (Mod S.O.O. 6219) – General data.

B. Adjust the Landing Lights

To adjust the landing lights for normal operation, proceed as follows:

1) Position aircraft on level ground.

2) Adjust setting of three nuts on mounting ring until light beam is parallel with aircraft center line (line of flight).

3) Adjust setting of nuts until light beam focuses its center on the ground at a point 140 feet ahead of pilot.

C. Operational Test Position Lights

1) With power on the buses, switch POSITION switch to ON and check that the left and right wing, and tail lights come on.

2) Switch POSITION switch to OFF.

D. Operational Test Beacon Lights

1) With power on the buses, switch on the BEACON switch to ON and check that the upper beacon lights come on.

2) Switch BEACON switch to OFF.

E. Operational Test Taxi Light

1) With power on the buses, switch TAXI switch to ON and check that taxi light comes on.

2) Switch TAXI switch to OFF.

F. Operational Test Wing Inspection Lights

1) With power on the buses, switch the WING INSP switch ON and check that right and left wing inspection lights come on.

2) Switch WING INSP switch to OFF.

G. Operational Test Strobe (Mod 6/2028)

WARNING HIGH VOLTAGE SHOCK HAZARD PRESENT IN STROBE LIGHT SYSTEM (PRE MOD 6/2447 ONLY).

1) With power on the buses, switch STROBE switch to ON and check that left and right wing strobe lights come on.

2) Switch STROBE switch to OFF.

8. Removal/Installation

WARNING FAILURE TO OBEY ALL THE SAFETY PRECAUTIONS WHEN DOING MAINTENANCE ON OR NEAR ELECTRICAL/ELECTRONIC EQUIPMENT MAY RESULT IN INJURY TO PERSONNEL AND/OR DAMAGE TO THE AIRCRAFT. REFER TO 24-00-00, ELECTRICAL POWER – GENERAL DATA, PARA 2.A.

A. Removal and Installation of the LED Taxi Light

1) General

These procedures are for the removal and installation of the LED taxi light assembly.

Equipment and Materials:

- Treating Solution, Chemical Conversion Coating (C1) & Epoxy Polyamide Primer (F19)
- Cleaning Solvent, Dynamold DS 108F (DS108)
- Sealing Compound, RTV 108 Clear Silicone

2) Removal of LED Taxi Light

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Open the TAXI LIGHT (B16) circuit breaker and fit safety clip.

c) Remove the tie wraps and clear heatshrink above the splices SP19 and SP227 at the nose wheel fork.

d) Disconnect the splices SP19 and SP227. Bag the open connections in dustless and moisture-free packaging and temporarily secure it away from the taxi light.

e) Remove and retain the washer, and nut securing the bracket to the nose wheel fork while securely holding the taxi light.

f) Remove and discard all the hardware attaching the taxi light to the bracket. Retain the bracket for installation.

g) Remove the taxi light assembly.

3) Installation of LED Taxi Light

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Discard the U-bracket that is supplied with the new light assembly and retain the hardware for attaching the bracket.

c) Reproduce a No. 40 drain hole in the bottom rim of light housing at 270 degrees. Treat the newly formed hole with the treating solution while ensuring drainage.

Note The angle is referencing from the mounting hole and clockwise as seen from the lens side.

d) Clean the original hole on the light housing located at 180 degrees from the mounting hole with the cleaning solvent. Plug the hole with the sealing compound.

e) Remove the four screws and the faceplate on the taxi light face and rotate the lens so the TOP mark on the lens is at 90 degrees. Reinstall the faceplate and the screws.

f) Install the taxi light assembly to the bracket with the hardware retained in Para 8.A.3).b).

g) Install the bracket to the fork with the washer and nut retained in Para 8.A.2).e). Torque the nut to 150 in-lbs.

Note Make sure the taxi light is aligned with the fork.

h) Attach a splice to each wire attached to the taxi light assembly. Slide in a 4-inch piece of clear heatshrink over one end of each splice. Connect them with the aircraft wiring at splicing points SP19 and SP227. Pull the heatshrinks over the splices and shrink them down.

i) Fold wires and tie wrap them together.

j) Remove the safety clip and close the TAXI LIGHT (B16) circuit breaker.

k) Remove all tools and materials from the work area.

l) Do an operational test of the taxi light system as per Para 7.E.